7.7 Answers: metric system and area

← Unit 7.7 · Practice

Check the units as well as the numbers: perimeter in cm or m, area in cm^2 or m^2. A right number with a missing unit is still a right number; just mention the unit.

Set 1 · Warm-up: lengths

1. 100 cm
2. 1 000 m
3. 10 mm
4. 300 cm
5. 2 000 m
6. 5 cm (one step up: : 10)
7. 4 m (two steps up: : 100)
8. 10 dm
9. 50 cm

Talk about it: smaller. Metres are bigger than centimetres, so you need fewer of them.

Set 2 · Building: converting with kommagetallen

1. 150 cm (m → cm is two steps down: × 100)
2. 275 cm
3. 3,5 km (three steps up: : 1 000)
4. 0,85 m (: 100)
5. 1 200 m (× 1 000)
6. 2,5 cm (: 10)
7. 40 cm (one step down: × 10)
8. 0,25 l (1 l = 1 000 ml)
9. 1 500 g
10. 0,75 kg
11. 300 g (1 ons = 100 g)
12. 1 000 ml

Talk about it: three steps (m → dam → hm → km), so : 1 000.

Set 3 · Building: perimeter

1. 16 cm (5 + 3 + 5 + 3)
2. 16 m (4 + 4 + 4 + 4)
3. 32 m
4. 8 m (2,5 + 1,5 + 2,5 + 1,5)
5. 5 cm (20 : 4)
6. 5 m (30 − 10 − 10 = 10 for the two short sides, so 5 each)
7. a. 40 m b. € 600 (40 × € 15)
8. 4 m

Set 4 · Building: area

1. 6 cm^2

2. 25 m²
3. 32 m²
4. 120 cm²
5. 14 m² (4 × 3 = 12, plus 4 × 0,5 = 2)
6. 4 cm (6 × 4 = 24)
7. 7 m (7 × 7 = 49)
8. € 240 (20 m² × € 12)
9. a. Neither: both are 36 m² b. B (A: 24 m, B: 26 m)
10. 100 m²

Talk about it: two shapes can have the same area but a different perimeter. They measure different things.

Set 5 · Stretch: composite shapes and m²/cm²

1. 20 m² (6 × 4 = 24, minus the bite 2 × 2 = 4)
2. 20 m (6 + 2 + 2 + 2 + 4 + 4; the same as the full rectangle)
3. 23 m² (15 + 8)
4. 74 m² (80 − 6)
5. 184 cm² (600 − 416)
6. 10 000 cm² (100 × 100)
7. 100 tiles (10 along each side)
8. 24 tiles (4 tiles per m², 6 m²; or 6 tiles along by 4 tiles across)
9. 6 m²

Talk about it: no. The two edges of the "bite" (2 m + 2 m) replace the two pieces of edge that were cut away (2 m + 2 m), so the walk around is just as long.

Puzzle corner

- With 12 m of fence: 1 × 5 (5 m²), 2 × 4 (8 m²), 3 × 3 (9 m²). The square, 3 × 3, gives the most space.
- With 20 m: 5 × 5 = 25 m² is best. The closer to a square, the more space.
- Against a wall (fence on three sides only): 3 m out from the wall and 6 m along it gives 18 m², the best with whole metres. Now the best shape is *not* a square; nice to discover.